

Arithmetic extracted from `riemann.F90`

FLASH Spark hydrodynamics

Scope. This document translates the executable arithmetic in `source/physics/Hydro/HydroMain/Spark/riemann.F90`. The common mathematics of `riemann` and `riemannOffloaded` is written once. The conversions inlined in the offloaded routine also occur in the included `prim2con.F90` and `prim2flx.F90`. The commented-out routine starting at line 736 is excluded.

1. Notation and input states

Let $s \in \{L, R\}$ denote the left or right state, and let n denote the normal coordinate selected by `dir`. For $j \in \{x, y, z\}$, $\delta_{jn} = 1$ when $j = n$ and is zero otherwise.

Symbol	Source variable or component
ρ_s, p_s	<code>V{L,R}(HY_DENS)</code> , <code>HY_PRES</code>
$\mathbf{u}_s, u_{n,s}$	<code>HY_VELX:HY_VELZ</code> , <code>velN{L,R}</code>
γ_s, ϵ_s	<code>HY_GAMC</code> , <code>HY_RHOE</code>
$\mathbf{B}_s, B_{n,s}, \psi_s$	<code>HY_MAGX:HY_MAGZ</code> , <code>magn{L,R}</code> , <code>HY_PSIB</code>
c_h, τ	<code>hy_C_hyp</code> , <code>hy_tiny</code>
U_s, F_s	<code>UL/UR</code> , <code>FL/FR</code>
U_s^*, F^*	<code>UCstarL/UCstarR</code> , <code>Fstar</code>
q^*, p^*	<code>qStar</code> , <code>pStar</code>

ϵ_s is the internal-energy density supplied in `HY_RHOE`; no ideal-gas substitution is made. Magnetic terms apply only when `SPARK_GLM` is defined; in hydrodynamics take $\mathbf{B}_s = \mathbf{0}$ and omit magnetic and ψ components.

The offloaded routine selects its states and shock flag as follows:

Source: `riemann.F90`, lines 334–340.

$$V_L = \text{uPlusArray}(1:\text{HY_NUM_VARS}, i_1 - 1, i_2, i_3), \quad (1)$$

$$V_R = \text{uMinusArray}(1:\text{HY_NUM_VARS}, i_1, i_2, i_3), \quad (2)$$

$$\text{inShock} = \bigvee_{k=i_1-1}^{i_1} [\text{shck}(k, i_2, i_3) \neq 0]. \quad (3)$$

F^* aliases `flux(:, i1, i2, i3)` in that routine. All dot products below are sums over the three coordinate components.

2. Sound speeds, GLM states, and outer waves

Source: *riemann.F90*, lines 82–158; 344–410. For each side,

$$a_s^2 = \frac{\gamma_s p_s}{\rho_s}, \quad c_{f,s} = \sqrt{a_s^2} \quad (\text{hydrodynamics}). \quad (4)$$

For GLM-MHD, the initial normal magnetic fields are retained in the subsequent Riemann arithmetic. The auxiliary GLM solution is

$$B_n^{\text{glm}} = \frac{B_{n,L} + B_{n,R}}{2} - \frac{\psi_R - \psi_L}{2c_h}, \quad (5)$$

$$\psi^{\text{glm}} = \frac{\psi_L + \psi_R}{2} - \frac{c_h}{2}(B_{n,R} - B_{n,L}). \quad (6)$$

The magnetic speed quantity and fast magnetosonic speed are

$$b_s^2 = \frac{\mathbf{B}_s \cdot \mathbf{B}_s}{\rho_s}, \quad (7)$$

$$c_{f,s} = \sqrt{\frac{1}{2} \left[a_s^2 + b_s^2 + \sqrt{(a_s^2 + b_s^2)^2 - \frac{4a_s^2 B_{n,s}^2}{\rho_s}} \right]}. \quad (8)$$

Before these square roots, the routine initializes $\text{ierr} = 0$ and applies the following checks in order:

$$\exists s : (0 < \rho_s < \tau) \vee (0 < p_s < \tau) \implies F^* = \mathbf{0} \quad \text{and return}, \quad (9)$$

$$(a_L^2 < 0) \vee (a_R^2 < 0) \implies \text{ierr} = 1 \quad \text{and return}. \quad (10)$$

These are the exact predicates in the source, rather than a general positivity test. The early returns precede the assignment of **speed**; the second return also precedes assignment of F^* .

The outer wave speeds and the returned maximum speed are

$$S_L = \min(u_{n,L} - c_{f,L}, u_{n,R} - c_{f,R}), \quad (11)$$

$$S_R = \max(u_{n,L} + c_{f,L}, u_{n,R} + c_{f,R}), \quad (12)$$

$$\text{speed} = \max(|S_L|, |S_R|). \quad (13)$$

The total pressure is

$$P_s = \begin{cases} p_s, & \text{hydrodynamics,} \\ p_s + \frac{1}{2} \mathbf{B}_s \cdot \mathbf{B}_s, & \text{GLM-MHD.} \end{cases} \quad (14)$$

3. Conservative states and physical fluxes

Source: *riemann.F90*, lines 162–165; 418–570. The scalar temporaries in the inlined conversions are

$$u_s^2 = \mathbf{u}_s \cdot \mathbf{u}_s = \sum_{j=x,y,z} u_{j,s}^2, \quad (15)$$

$$B_s^2 = \sum_{j=x,y,z} B_{j,s}^2, \quad (UB)_s = \sum_{j=x,y,z} u_{j,s} B_{j,s}, \quad (16)$$

$$E_s = \frac{1}{2} \rho_s u_s^2 + \epsilon_s + \frac{1}{2} B_s^2. \quad (17)$$

For hydrodynamics, $B_s^2 = 0$ and $E_s = \frac{1}{2} \rho_s u_s^2 + \epsilon_s$. The assigned conservative components are

$$(U_s)_\rho = \rho_s, \quad (U_s)_{m_j} = \rho_s u_{j,s}, \quad (U_s)_E = E_s, \quad (18)$$

$$(U_s)_{B_j} = B_{j,s}, \quad (U_s)_\psi = \psi_s \quad (\text{GLM-MHD only}). \quad (19)$$

The three directional **select case** blocks are represented exactly by the component formulas below, with $n = x, y$, or z .

Hydrodynamics:

$$(F_s)_\rho = \rho_s u_{n,s}, \quad (20)$$

$$(F_s)_{m_j} = \rho_s u_{n,s} u_{j,s} + p_s \delta_{jn}, \quad (21)$$

$$(F_s)_E = (E_s + p_s) u_{n,s}. \quad (22)$$

GLM-MHD:

$$(F_s)_\rho = \rho_s u_{n,s}, \quad (23)$$

$$(F_s)_{m_j} = \rho_s u_{n,s} u_{j,s} - B_{n,s} B_{j,s} + P_s \delta_{jn}, \quad (24)$$

$$(F_s)_E = (E_s + P_s) u_{n,s} - B_{n,s} (\mathbf{u}_s \cdot \mathbf{B}_s), \quad (25)$$

$$(F_s)_{B_j} = u_{n,s} B_{j,s} - u_{j,s} B_{n,s}, \quad (26)$$

$$(F_s)_{B_n} = 0, \quad (F_s)_\psi = 0. \quad (27)$$

The ψ flux is zero at this stage because the flux vectors are initialized to zero; final GLM fluxes overwrite it later.

Initialization detail. The helper `prim2con` zeroes its whole output for each call. The offloaded inline code zeroes U_L , but does not zero the whole U_R before assigning the components above. Both inline physical flux vectors are zeroed before component assignment.

4. HLL state and contact wave

Source: *riemann.F90*, lines 168–228; 575–639. The intermediate HLL conservative vector is

$$U^{\text{hll}} = \begin{cases} U_L, & S_L > 0, \\ \frac{S_R U_R - S_L U_L - F_R + F_L}{S_R - S_L}, & S_L \leq 0 \text{ and } S_R \geq 0, \\ U_R, & \text{otherwise.} \end{cases} \quad (28)$$

For MHD define the quantities actually read from that vector:

$$\mathbf{B}^* = \mathbf{B}^{\text{hll}}, \quad B_n^{\text{hll}} = B_n^*, \quad \mathbf{u}^{\text{hll}} = \frac{\mathbf{m}^{\text{hll}}}{\rho^{\text{hll}}}. \quad (29)$$

The contact speed is

$$q^* = \frac{\rho_R u_{n,R} (S_R - u_{n,R}) - \rho_L u_{n,L} (S_L - u_{n,L}) + P_L - P_R - B_{n,L}^2 + B_{n,R}^2}{\rho_R (S_R - u_{n,R}) - \rho_L (S_L - u_{n,L})}. \quad (30)$$

In hydrodynamics the magnetic terms are zero. Introduce the source temporaries **numerL/R** and **denomL/R**:

$$N_s = S_s - u_{n,s}, \quad D_s = S_s - q^*, \quad s \in \{L, R\}. \quad (31)$$

The star pressure is calculated from the left state:

$$p^* = \begin{cases} \rho_L N_L (q^* - u_{n,L}) + P_L, & \text{hydrodynamics,} \\ \rho_L N_L (q^* - u_{n,L}) + P_L - B_{n,L}^2 + (B_n^{\text{hll}})^2, & \text{GLM-MHD.} \end{cases} \quad (32)$$

The two star densities and energies are

$$\rho_s^* = \frac{\rho_s N_s}{D_s}, \quad (33)$$

$$E_s^* = \frac{E_s N_s + p^* q^* - P_s u_{n,s}}{D_s} \quad (\text{hydrodynamics}), \quad (34)$$

$$E_s^* = \frac{E_s N_s + p^* q^* - P_s u_{n,s}}{D_s} - \frac{B_n^{\text{hll}} (\mathbf{B}^* \cdot \mathbf{u}^{\text{hll}}) - B_{n,s} (\mathbf{B}_s \cdot \mathbf{u}_s)}{D_s} \quad (\text{GLM-MHD}). \quad (35)$$

The magnetic fields used here are the original input fields and the HLL fields, not the auxiliary B_n^{glm} . The latter is used only for the final GLM flux overwrite.

5. Star momenta and final numerical flux

Source: *riemann.F90*, lines 230–304; 642–723. For each side, the assigned star components satisfy

$$(U_s^*)_\rho = \rho_s^*, \quad (U_s^*)_E = E_s^*, \quad (36)$$

$$(U_s^*)_{m_n} = \rho_s^* q^*. \quad (37)$$

For the two tangential coordinates $t \neq n$,

$$(U_s^*)_{m_t} = \begin{cases} \frac{\rho_s u_{t,s} N_s}{D_s}, & \text{hydrodynamics,} \\ \frac{\rho_s u_{t,s} N_s - B_n^* B_t^* + B_{n,s} B_{t,s}}{D_s}, & \text{GLM-MHD.} \end{cases} \quad (38)$$

For MHD, $(U_s^*)_{B_j} = B_j^*$ for $j = x, y, z$. Thus $n = x$ uses $t = y, z$; $n = y$ uses $t = x, z$; and $n = z$ uses $t = x, y$.

The HLLC flux follows the source's ordered branches:

$$F^{\text{HLLC}} = \begin{cases} F_L, & S_L \geq 0, \\ F_L + S_L(U_L^* - U_L), & S_L < 0 \text{ and } q^* \geq 0, \\ F_R + S_R(U_R^* - U_R), & S_L < 0, q^* < 0, S_R \geq 0, \\ F_R, & \text{otherwise.} \end{cases} \quad (39)$$

If `hy_hybridRiemann` $\wedge$ `inShock` is true, this is overwritten by the HLLE flux:

$$F^{\text{HLLE}} = \begin{cases} F_L, & S_L > 0, \\ F_R, & S_L \leq 0 \text{ and } S_R < 0, \\ \frac{S_R F_L - S_L F_R + S_R S_L (U_R - U_L)}{S_R - S_L}, & \text{otherwise.} \end{cases} \quad (40)$$

$$F^* \leftarrow \begin{cases} F^{\text{HLLE}}, & \text{hy_hybridRiemann} \wedge \text{inShock}, \\ F^{\text{HLLC}}, & \text{otherwise.} \end{cases} \quad (41)$$

Finally, for GLM-MHD, two components are overwritten regardless of which solver supplied the other components:

$$\boxed{(F^*)_{B_n} = \psi^{\text{glm}}, \quad (F^*)_\psi = c_h^2 B_n^{\text{glm}}.} \quad (42)$$

Implementation limits. Neither routine assigns the star ψ component or initializes the entire star vectors before vector flux arithmetic. The final ψ flux is overwritten as shown above. These equations describe the explicitly assigned components; no missing assignments or denominator guards have been added to the algorithm.